

Bachelor Project Group, Master Seminar & Lab in Visual Computing **Organization**

Winter Term 2026/27

Prof. Dr. Reinhard Klein

Prof. Dr. Thomas Schultz

Prof. Dr. Matthias Hullin

Prof. Dr. Björn Krüger



Institute of Computer Science II
Visual Computing Department



Bachelor Courses

Project Groups: (Bachelor Informatik)

- BA-INF 051 – Computergraphik
- BA-INF 051 – Digital Material Appearance

Master Courses

Master Labs: (Master Comp. Sc.)

- MA-INF 2308 – Computer Graphics
- MA-INF 2311 – Computer Animation
- MA-INF 2316 – Digital Material Appearance

Master Seminars: (Master Comp. Sc.)

- MA-INF 2207 – Computer Graphics
- MA-INF 2210 – Computer Animation
- MA-INF 2215 – Digital Material Appearance
- MA-INF 2219 – Visualization and Medical Image Analysis

Course Overview

	Project Group	Master Seminar	Master Lab
Seminar Part	✓	✓	✓
Practical Part	✓	✗	✓

Seminar Part:

- Read and understand provided literature.
- At the end of the semester:
 - Written report (graded),
 - Final talk (graded).

Practical Part:

- *Implementation* of techniques discussed in provided literature.
- At the end of the semester:
 - Software documentation in written report (graded),
 - Software demo during final talk (graded).

Supervision

- Each topic is supervised by one member of the research staff.
- Report your progress in meetings.
- Milestones will be provided by the supervisor:
 - E.g. You should have read the provided paper after one month.
- Your supervisor will be available for all questions concerning your project.

Seminar Talk

- Present your work at the end of the semester:
 - 30 minutes talk + 15 minutes discussion.
- Slides for the talk:
 - PowerPoint or PDF slides.
 - Discuss the structure with your supervisor before you start.
 - Trial talk is always a nice option to see if the talk matches the expectations.

Written Report

- Hand in **1 week** before the seminar talk!
- Written in LaTeX:
 - Template available at <https://cg.cs.uni-bonn.de/courses>.
 - Around 10 pages in length.
- Sound and comprehensible summary of provided literature in your own words.
- Adhere to scientific standards:
 - Provide proof for your statements.
 - Use references to make your sources transparent.
 - Distinguish your contributions from the work of others.
 - Do not be biased.
 - Mark quotes as such.

Important Dates

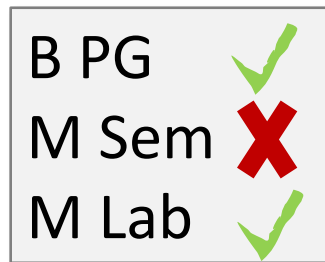
- Ask your supervisor for feedback early!
- Register in Basis **after** talking to your supervisor.
- Deadline for registration in Basis:

31. October 2026

- Final report and slides are due **one week** before the talk.
- Seminar talks:
 - All talks for one module take place on the same day.
 - Talks will take place during the lecture-free period.
 - Date for talks will be announced later during the semester.
 - All participating students have to attend all talks for their particular course!

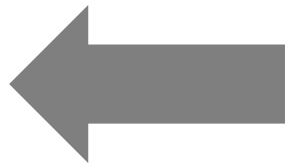
Topic Selection

- Interested in a topic?
 - Contact the supervisor by e-mail or after the presentation!
 - E-mail addresses and names are provided on the slide(s) of the respective topic.
 - If in doubt, contact Jan Spindler (jspindle@cs.uni-bonn.de)



**These symbols tell you the
courses a topic is suitable for**

- Slides for organization and topics are available on our homepage:
 - <https://cg.cs.uni-bonn.de/teaching/ws-2026-27-ma-inf-2308-master-lab-graphics>
 - Username: **cglab26ws**
 - **Password:** **PixelNinja**



Schedule for Topics

- Topics will now be presented in this order:
 - Visualization and Medical Image Analysis
 - Digital Material Appearance
 - Computer Graphics
 - Computer Animation

Visualization and Medical Image Analysis

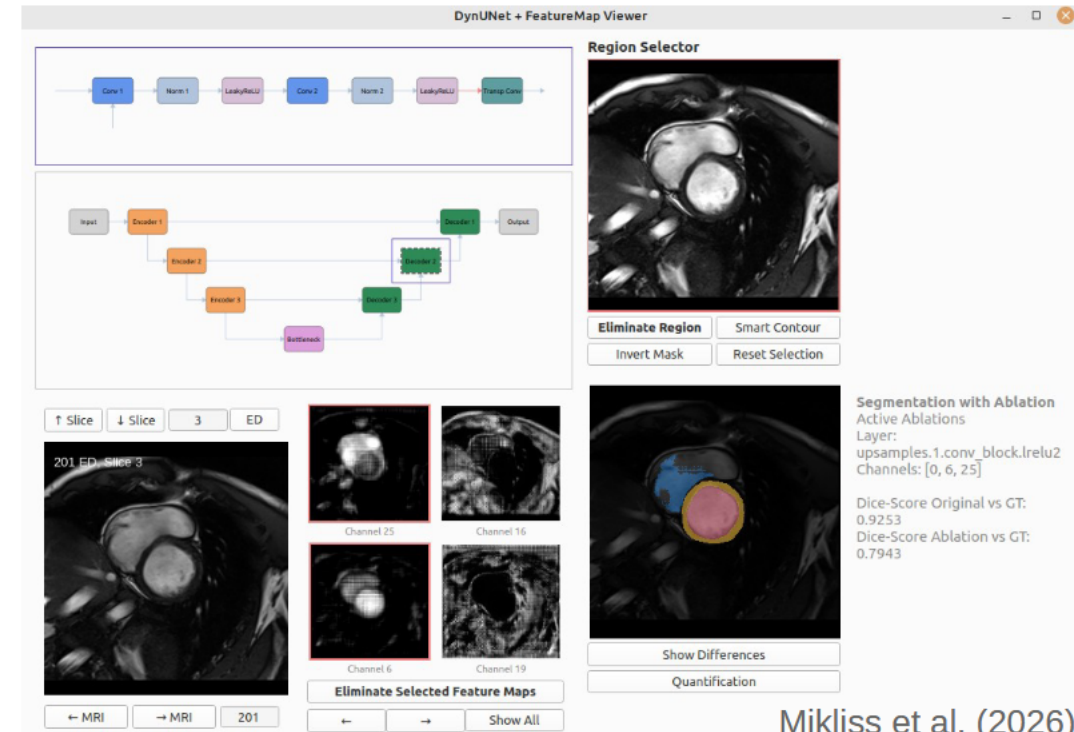
Prof. Dr. Thomas Schultz

- Mail: mikliss@cs.uni-bonn.de or schultz@cs.uni-bonn.de
- Alternative B-IT presentation on 15.10.2026 at 13:00 in 0.107

Interactive Feature Map Ablation in CNNs for Image Segmentation

Annika Mikliss (mickliss@cs.uni-bonn.de)

- CNNs achieve their results through **complex nonlinear interactions**.
- Our **interactive feature map ablation** tool supports the exploration of CNN **segmentation strategies** through functionalities such as different sorting strategies, gradient visualization and region-specific ablations.
- We focused on a U-Net architecture applied to a cardiac imaging datasets.
- **Tasks:**
 - Make the tool more flexible and usable for **other architectures** and datasets.
 - Further **feature development** to guide feature map exploration.
 - Perform **experiments** to find out more about segmentation strategies.

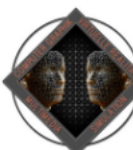


Mickliss et al. (2026)
Revealing Learned Medical Image Segmentation Strategies
through Interactive Feature Map Ablation in CNNs
<https://cg.cs.uni-bonn.de/publications/mickliss-2026-revealing>

B PG	✗
M Sem	✗
M Lab	✓



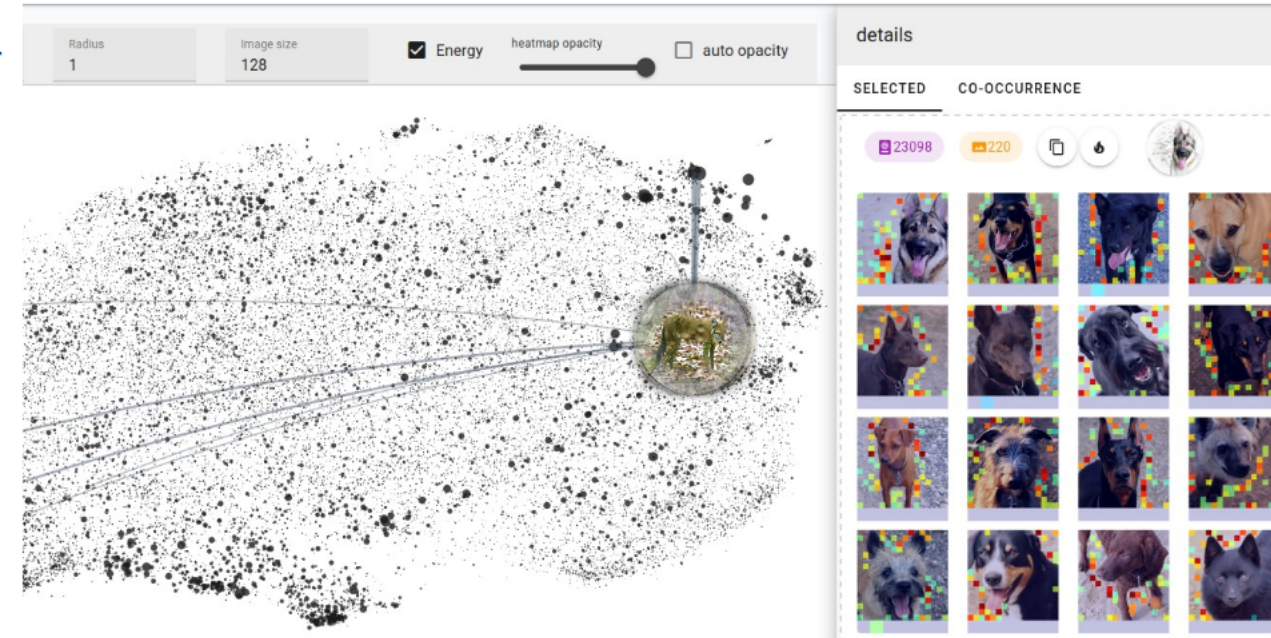
UNIVERSITÄT **BONN**



Concept Exploration of SAM

Annika Mikliss (mickliss@cs.uni-bonn.de)

- **Foundation models** such as DINO and SAM impress with their outstanding performance – though they are opaque.
- Fel et al. investigated DINOv2's **internal representations** by encoding them using **sparse autoencoders**.
- In their DinoVision demo they make the extracted **concept vectors** explorable. We can inspect on which images the concept activates most strongly and explore related concepts.
- **Tasks:**
 - Train a **sparse autoencoder** and extract the concept vectors of **SAM**.
 - Explore and analyze the extracted concepts.



Fel et al. (2026)
Into the Rabbit Hull: From Task-Relevant Concepts in DINO to
Minkowski Geometry
<https://arxiv.org/pdf/2510.08638>

B PG	✗
M Sem	✗
M Lab	✓

Digital Material Appearance

Prof. Dr. Matthias Hullin

Streaming & Compressing 4D Gaussian Splatting

Stefan Schulz (sschulz@cs.uni-bonn.de)

- Papers:

- Temporal Smoothness-Aware Rate-Distortion Optimized 4D Gaussian Splatting (Lee et al.)
- Optimized Minimal 4D Gaussian Splatting (Lee et al.)
- QUEEN: Quantized Efficient encoding of Dynamic Gaussians for Streaming Free-viewpoint Videos (Girish et al.)
- Compact 3D Gaussian Representation for Radiance Field (Lee et al.)
- MEGA: Memory-Efficient 4D Gaussian Splatting for Dynamic Scenes
- AirGS: Real-Time 4D Gaussian Streaming for Free-Viewpoint Video Experiences (Wang et al.)
- PD-4DGS: Progressive Decomposition of 4D Gaussian Splatting for Bandwidth-Adaptive Dynamic Scene Streaming (Li et al.)
- Feel free to come up with your own proposals :D

B PG	✗
M Sem	✓
M Lab	✗

Clouds Simulation

Fernando Edelstein (edelstein@cs.uni-bonn.de)

Motivation: Realistic cloud formations and atmospheric effects have been a research focus for years in order to create immersive video games, flight simulators, and virtual production.

Problem: Accurately simulating the complex interactions of light scattering through dense volumes requires. Achieving this in real-time requires balancing physical accuracy with performance, relying on approximations like volumetric ray marching, procedural noise generation, and customised lighting models.

Your Task: Optimize the rendering pipeline using compute shaders, temporal upscaling (reprojection), or half-resolution rendering techniques to maintain strict real-time performance constraints.

Background: OpenGL, Mitsuba, Vulkan, Computer Graphics



B PG	✗
M Sem	✗
M Lab	✓

Heuristic Algorithms Global Illumination

Fernando Edelstein (edelstein@cs.uni-bonn.de)

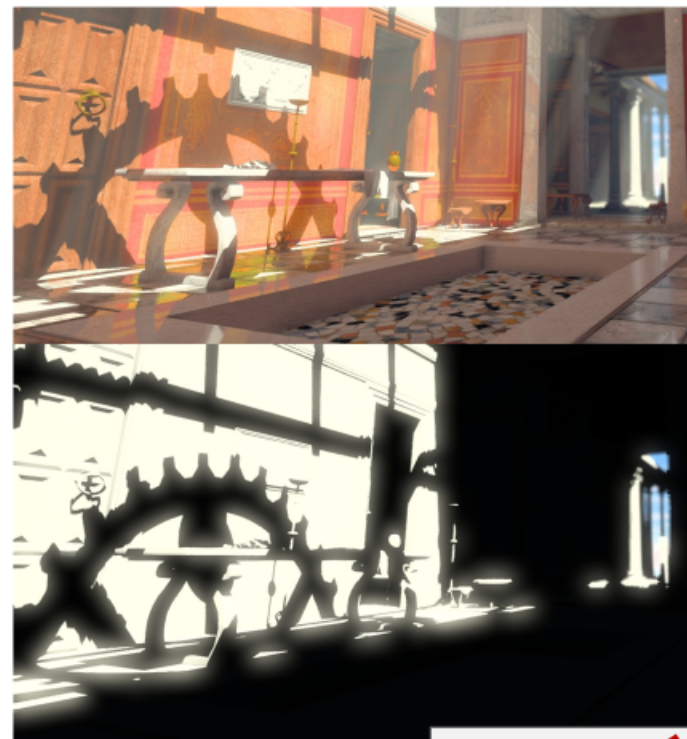
Motivation: Real-time global illumination is essential for photorealistic rendering, but traditional probe-based systems like DDGI rely on rigid, dense spatial grids.

Problem: These grids waste significant computational resources on empty space and frequently suffer from light leaking across thin geometry.

Your Task: Instead of pre-allocating a massive grid, use a Screen Space approach, take advantage of Heuristic algorithms for probe placement.

Develop an interpolation scheme to gather indirect lighting from these irregularly placed probes during the final shading pass, ensuring temporal stability and minimizing light leaking across geometry boundaries.

Background: Computer Graphics, OpenGL, Mitsuba, Vulkan, Ray Tracing



B PG	✗
M Sem	✗
M Lab	✓

Computer Graphics

Prof. Dr. Reinhard Klein

Deep Learning for Weather Forecasting

Nils Wandel (wandel@cs.uni-bonn.de)

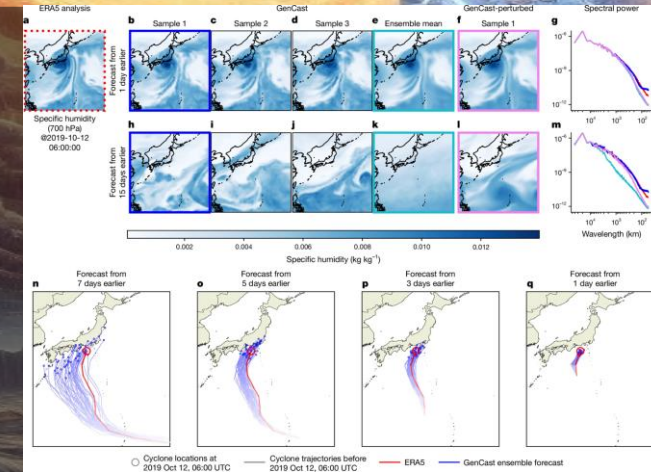
Seminar:

“Probabilistic weather forecasting with machine learning”

Ilan Price et al. (2024)

Tasks:

- Understand and explain the underlying theoretical concepts and methodology
- Read and compare the articles that are covered in this paper
- Write a report and give a presentation on this topic



Lab / Projektgruppe:

“Deep Learning for weather forecasting on ERA5 dataset”

Tasks:

- Install and run a deep-learning framework
- Try out different spherical neural network architectures and compare performance on ERA5 dataset
- Write a report and give a presentation about your findings

FULL, FREE AND OPEN
ACCESS TO DATA

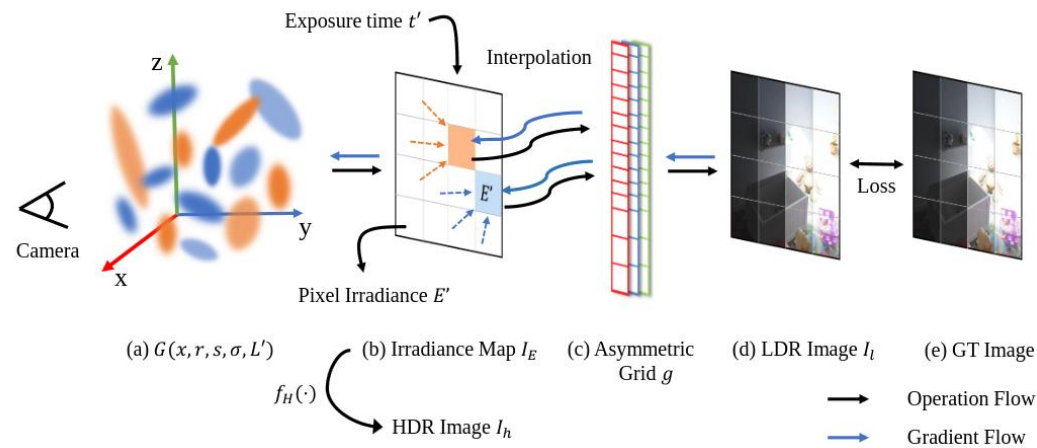


- ATMOSPHERE MONITORING
- MARINE ENVIRONMENT MONITORING
- LAND MONITORING
- CLIMATE CHANGE
- EMERGENCY MANAGEMENT
- SECURITY

HDRGS for Avatar Reconstruction

Simon Katz (katz@cs.uni-bonn.de)

- Goal: Multiview-video reconstruction of humans (via 3dgs) and camera response function (CRF)
- Problem: Datasets are usually distributed as LDR images. HDR reconstruction is thus difficult, and unknown HDR $\rightarrow$ SDR mapping can complicate reconstruction
- Task: implement CRF training into existing Avatar pipeline (PiG-Avatar) and evaluate it. Alternatively, for a Seminar, evaluate current approaches to avoiding camera artifacts.
- Relevant works: HDRGS (, PPISP)
- Tools: Python, torch



B PG	✗
M Sem	✓
M Lab	✓

Single View Avatar Reconstruction

Simon Katz (katz@cs.uni-bonn.de)

- Problem: Generating high-quality Avatars usually requires multiview input, which can be prohibitive for many applications
- Task: Evaluate current works that extend Avatar reconstruction to monocular input, and categorize their methods.
- Relevant works: AniGS, IDOL, LHM



B PG	✗
M Sem	✓
M Lab	✗

Capturing Scene-Aware Human Motion in VR

Sina Kitz (kitz@cs.uni-bonn.de)

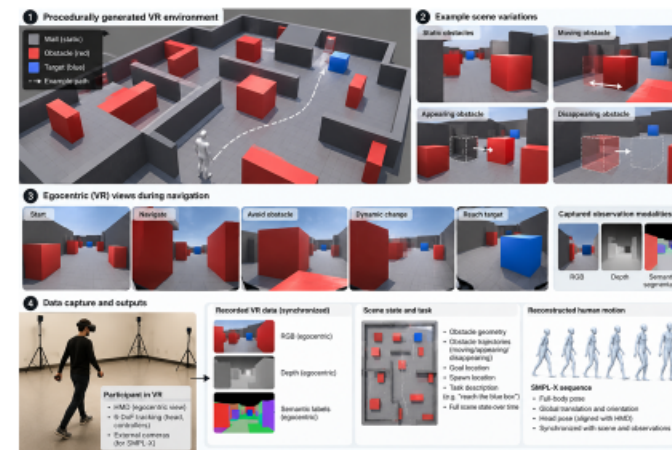
- Current **human-motion datasets** have only limited paired **human $\leftrightarrow$ scene** interactions.
- need scalable datasets combining full-body SMPL-X motion, egocentric observations, navigation goals, and temporally aligned scene state, especially for **dynamic environments**.
- Moving Through Clutter**: “Learning Scene-Aware Humanoid Locomotion through 3D Clutter from Immersive Human Demonstrations” by Wang et al, (Sept. 2026)

Dataset Pipeline

- Participants navigate procedurally generated **dynamic** VR environments
- simple tasks such as reaching a target while avoiding obstacles.
- records the complete scene state, task, egocentric observations, and VR tracking data, while external video is used to reconstruct synchronized SMPL-X motion.

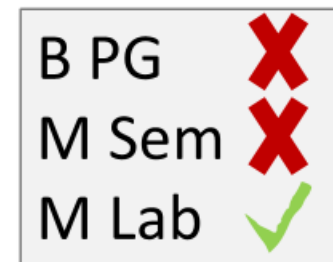
Evaluation

train a goal-conditioned navigation policy to predict human trajectories in static and dynamic scenes.



This seminar is for you if:

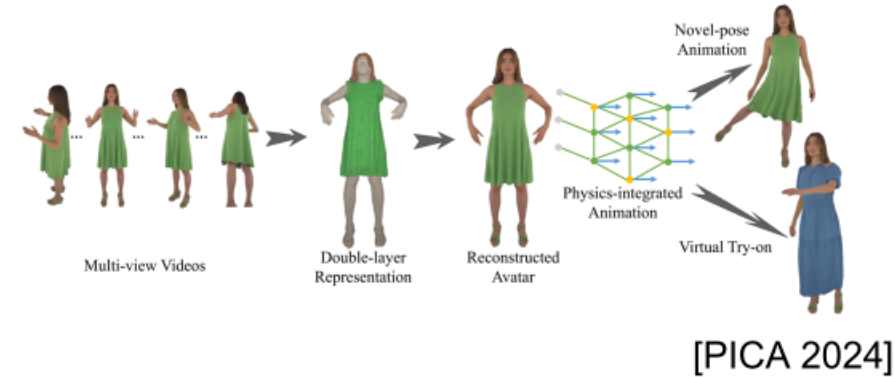
- you are interested in human motion, VR and ML
- enjoy working as a team
- familiarity with Unreal engine (helpful, not required)



Deformable Garments for Gaussian Avatars

Sina Kitz (kitz@cs.uni-bonn.de)

- Gaussian avatars can reconstruct highly detailed clothing, but loose garments exhibit secondary motion, inertia, collisions, and deformation that are hard to determine from the current body pose alone.
- **How can Gaussian avatars represent clothing that deforms plausibly under novel motion and physical interaction?**
 - **SMAGA:** Secondary Motion-Aware 3D Clothed Gaussian Avatars from Monocular Videos — Seungeun Lee et al., 2026 (ICLR)
 - **PICA:** Physics-Integrated Clothed Avatar — Bo Peng et al., 2025 (IEEE TVCG)
 - **MPMAvatar:** Learning 3D Gaussian Avatars with Accurate and Robust Physics-Based Dynamics — Changmin Lee et al., 2025 (NeurIPS)
 - **DAMA:** Disentangled Body-Anchored Gaussians for Controllable Multi-Layered Avatars — Daniel Eskandar et al., 2026 (CVPR Workshops)



This seminar is for you if:

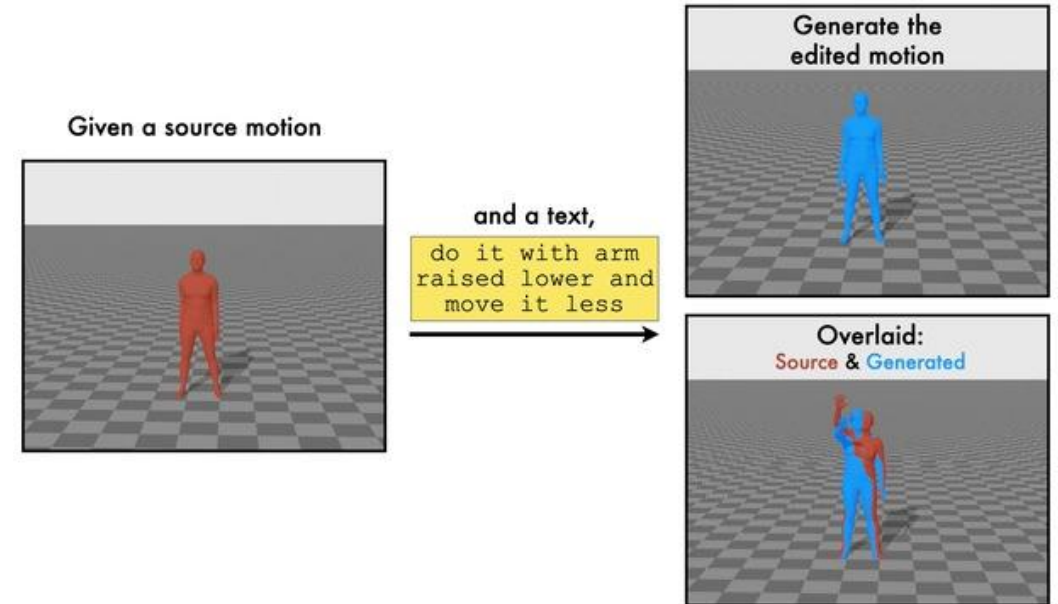
- you are not scared of some physics & maths
- you are interested in simulation & ML
- familiarity with 3D Gaussian Splatting is a plus

B PG	✗
M Sem	✓
M Lab	✗

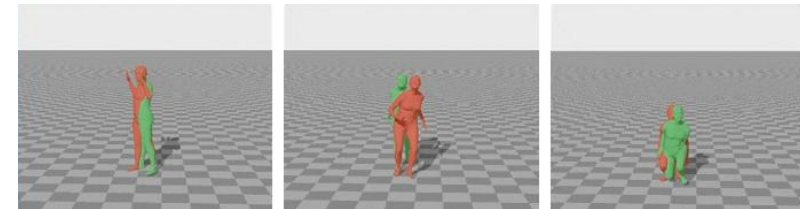
Text-Driven Human Motion Editing

Jan Spindler (jspindle@cs.uni-bonn.de)

- **Motivation:** AI models can now turn a text prompt into realistic 3D human motion. Results are rarely perfect on the first try, so animators want to fix them in plain words, like “do it in a smaller circle.”
- **Problem:** An edit must change exactly what the text asks and keep everything else intact. Recent methods bet on different fixes: more training data, finer control over which joints change, or larger pretrained models.
- **Your task:** Study four key papers (2024 to 2026), compare them on a shared benchmark, and work out which bet the evidence supports and what is still open.
- **Background:** Basic deep learning is enough. Diffusion models, transformers and the MotionFix benchmark are learned along the way.



Source motion + edit text → edited motion (red: source, blue: edited). Source: MotionFix, SIGGRAPH Asia 2024



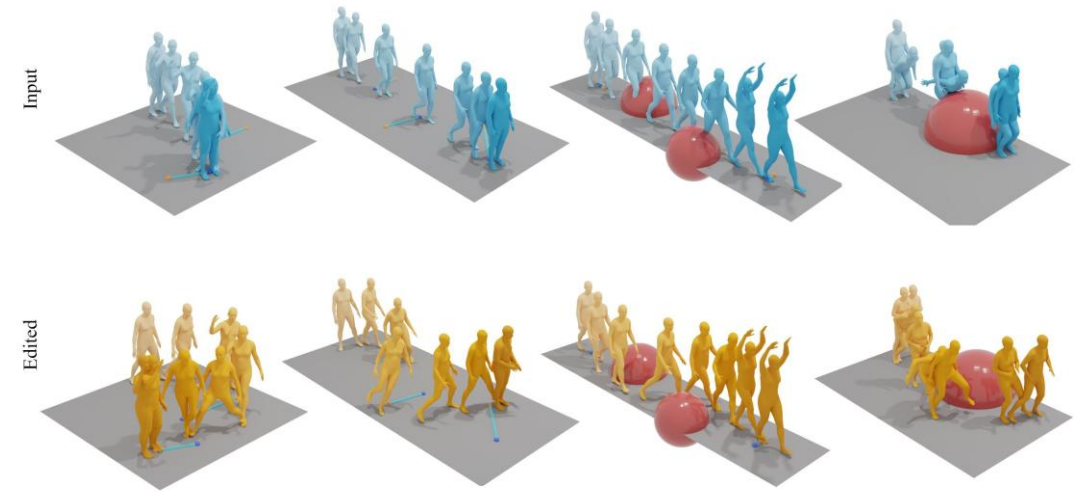
Real edit texts from the dataset: “tilt to the opposite side first”, “do the same with the opposite leg”, “use the opposite leg to get up” (red: source, green: target). Source: MotionFix

B PG	✗
M Sem	✓
M Lab	✗

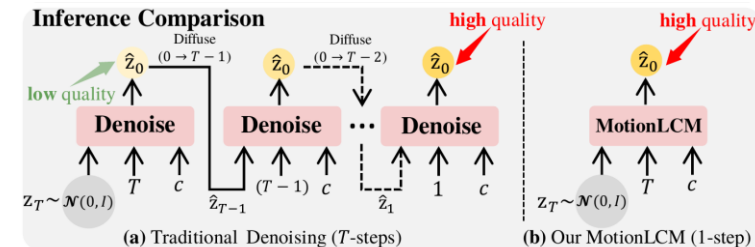
Fast Noise Optimization for Human Motion

Jan Spindler (jspindle@cs.uni-bonn.de)

- **Motivation:** Diffusion models turn a text prompt into realistic 3D human motion. Optimizing their input noise steers the result toward exact goals, like reaching a spot or avoiding an obstacle.
- **Problem:** This optimization is slow. A distilled one-step “student” model makes it fast, but does noise tuned on the student still work for the original “teacher” model?
- **Your task:** Build the student-to-teacher pipeline, measure where the two models agree, and fix where they drift apart.
- **Tools:** Python, PyTorch and diffusion model basics. Builds on public code (MotionLCM, DNO) and the HumanML3D dataset.



Noise optimization edits a motion to meet new goals (top: input, bottom: edited). Source: DNO, CVPR 2024



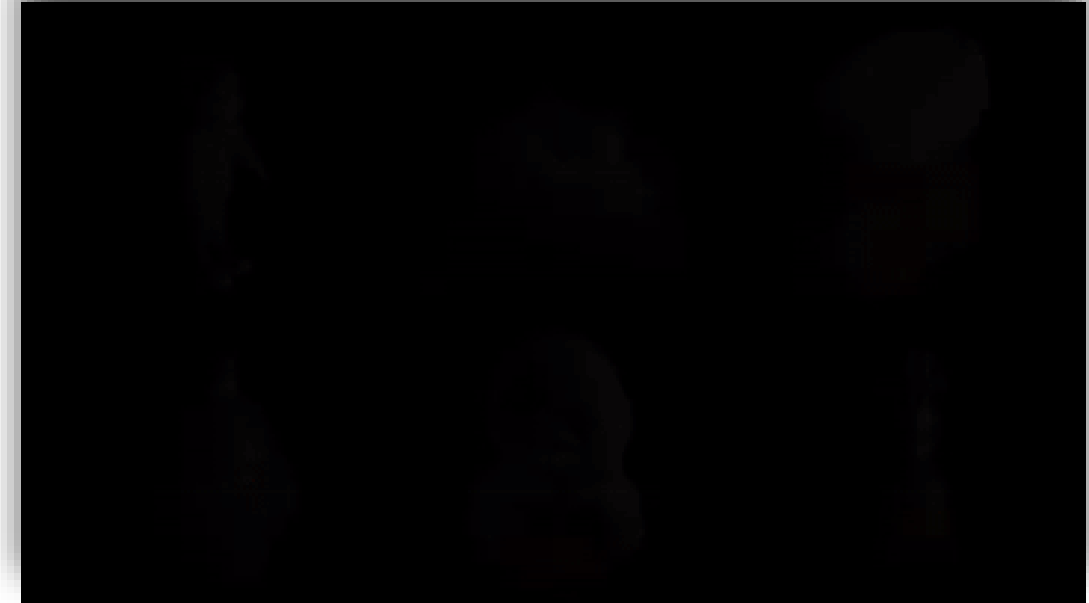
Teacher: many denoising steps. Student: one step. Source: MotionLCM-V2

B PG	✗
M Sem	✗
M Lab	✓

Relighting of Dynamic 3D Gaussian Models

Jan Müller (muellerj@cs.uni-bonn.de) **Contact by 12.10.2026 due to absence**

- Standard 3D Gaussian Splatting (3DGS) reconstruction often bakes in lighting. AR/VR Application, VFX, and robotics require dynamic real-world scenes to be seamlessly inserted into new lighting environments.
- Underconstrained Problem:
 - What happens when we only have monocular video or a collection of uncalibrated photos?
 - Disentangling motion, geometry, material properties (BRDF), and complex lighting is a highly ill-posed, inverse rendering problem.
- **Goal:** Exploring recent 3DGS methods for outdoor scenes, dynamic full-body avatars, and high-fidelity faces.
- Explore how current research approaches underconstrained setups using:
 - Physics-based priors and shading models.
 - Parametric models (e.g., SMPL for bodies) to guide motion and geometry.
 - Signed Distance Function (SDF) regularization.



GS³: Efficient Relighting with Triple Gaussian Splatting - Bi et al.

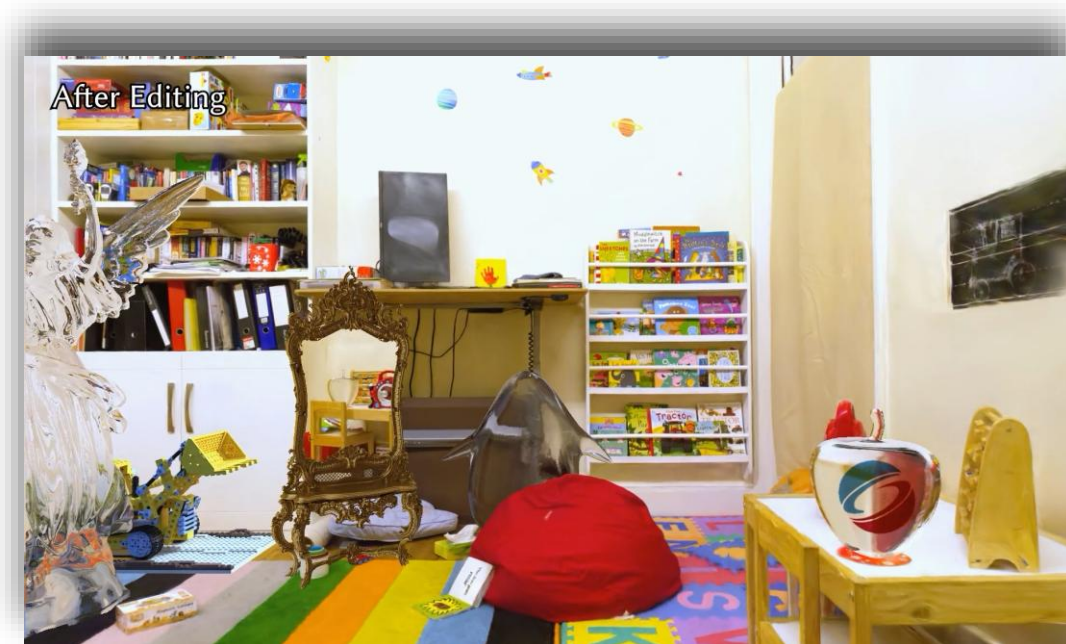
B PG	✗
M Sem	✓
M Lab	✗



Refraction and Reflection in 3D Gaussian Models

Jan Müller (muellerj@cs.uni-bonn.de) **Contact by 12.10.2026 due to absence**

- Standard 3D Gaussian Splatting relies on Spherical Harmonics (SH) and rasterisation. It is incredibly fast but fundamentally fails at capturing sharp reflections, mirrors, and the bending of light at surface boundaries (refraction).
- Goal: Exploring extensions of 3DGS which explicitly handle aspects of complex light transport.
- Provide overview over different approaches to solve the problem:
 - Shading & View-Dependent Approximations: Environment maps, screen-space approximations.
 - Primitive Extensions: Mirrored scene geometry
 - Ray-Tracing: Calculating ray-Gaussian intersections.
 - Hybrid Geometry Methods (SDFs + Gaussians): Constructing stable normals from proxy geometry.



TransparentGS: Fast Inverse Rendering of Transparent Objects with Gaussians - Huang et al.

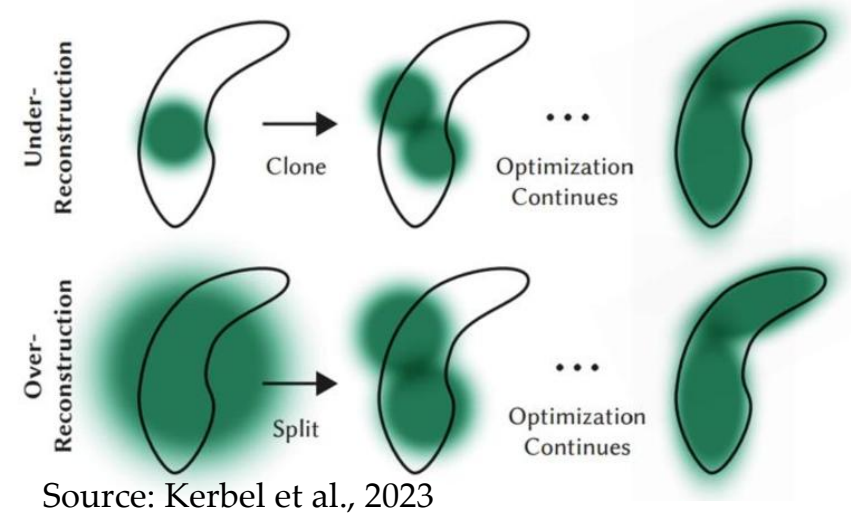
B PG	✗
M Sem	✓
M Lab	✗



Continuous Densification in 3D Gaussian Splatting.

Jan Müller (muellerj@cs.uni-bonn.de) **Contact by 12.10.2026 due to absence**

- 3D Gaussian Splatting (3DGS) reconstructs scenes from images which requires to allocate geometry to the correct parts of the scenes:
 - New particles are created by duplicating or dividing the existing scene geometry at fixed intervals during the trainings
 - The decision to densify is based on a coarse heuristic which can lead to errors in the reconstruction.
- **Questions we want to investigate:**
 - Can we move away from densification at fixed intervals?
 - What heuristic is best suited to this continuous densification?
- **Prerequisites:** Familiarity with Python, Numpy or PyTorch
 - Not afraid of C/C++
- If you want to look at related work relevant to this topic:
 - Steepest Descent Density Control for Compact 3D Gaussian Splatting, Wang et al., 2025
 - Faster 3D Gaussian Splatting Convergence via Structure-Aware Densification, Lyu et al., 2026



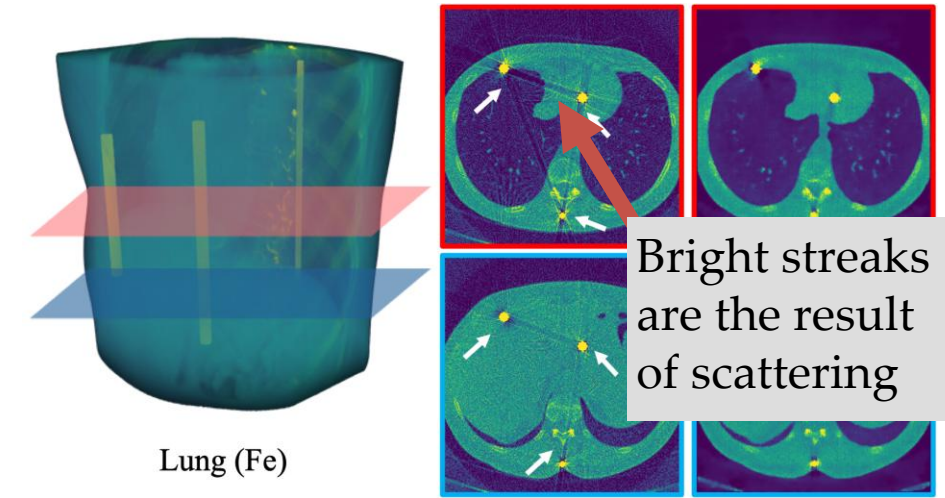
B PG	✓
M Sem	✗
M Lab	✓



Metal Scatter Artifact Reduction in CT Reconstruction

Jan Müller (muellerj@cs.uni-bonn.de) **Contact by 12.10.2026 due to absence**

- When metal objects are scanned using X-rays, some of the high energy particles scatter within the object.
 - This is inconsistent with assumptions made by popular CT-reconstruction methods that assume:
 - Attenuation according to Beer's law.
 - Uniformly distributed energy distribution of the observed photons.
- **Questions we want to investigate:**
 - Evaluate existing methods on a novel dataset with extreme scattering.
 - Investigate statistical methods to further reduce the scattering amount.
- **Prerequisites:** Familiarity with Python, Numpy or PyTorch
 - Not afraid of C/C++
- If you want to look at related work relevant to this topic:
 - R2-Gaussian: Rectifying Radiative Gaussian Splatting for Tomographic Reconstruction, Zha et al., 2024
 - Splat-based Metal Artifact Reduction in Cone-Beam CT via Polychromatic Modeling, Choi et al., 2025



Source: Choi et al., 2023

B PG	✗
M Sem	✗
M Lab	✓





- • • Klinik und Poliklinik
- • • für Epileptologie
- • • Universitätsklinikum Bonn



Master Seminar & Lab (Winter Term 2026/2027)

Computeranimation | Human-Centered Sensing and Digital Health

Prof. Dr. Björn Krüger // bkrueger@uni-bonn.de

Bonn, 06. Oktober 2026

www.epileptologie-bonn.de



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Master Seminar & Lab

Automatic Seizure Event Boundary Detection in Clinical EEG

AG Krüger (Nazila.AhmadiDaryakenari@ukbonn.de)

Accurate seizure onset and offset annotations are important for training and evaluating seizure detection systems, but clinical EEG annotations may be incomplete or imprecise.

- **Research question:** “Can seizure onset and offset be automatically refined from clinical EEG signals?”
- **Lab:** Develop and compare methods for identifying seizure boundaries from EEG, using window-level predictions, temporal smoothing, change-point detection, or lightweight temporal models.
- **Tools:** Python, PyTorch

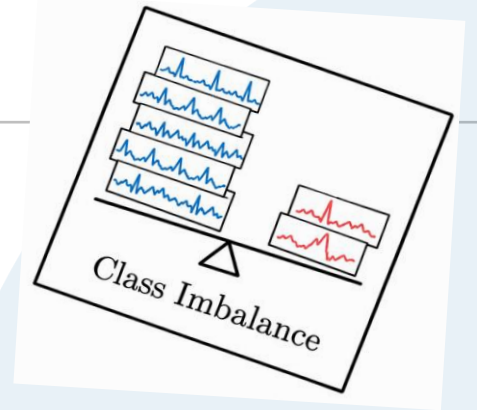
B PG	✗
M Sem	✗
M Lab	✓



Master Seminar & Lab

Class Imbalance Handling Methods for Physiological Time-Series

AG Krüger (Nazila.AhmadiDaryakenari@ukbonn.de)



Physiological time-series datasets often contain strongly imbalanced classes, particularly when detecting rare clinical events such as seizures, arrhythmias, apnea episodes, or other abnormal physiological conditions.

- **Seminar:** Write a scoping review of methods for handling class imbalance in physiological time-series classification, comparing sampling, augmentation, loss-based, and synthetic data approaches across different physiological signals.
- **Tools:** Scientific Research

B PG	✗
M Sem	✓
M Lab	✗



Master Seminar & Lab

Data Augmentation Methods for Physiological Time-Series Classification



AG Krüger (Nazila.AhmadiDaryakenari@ukbonn.de)

Physiological time-series datasets are often limited in size and can contain highly imbalanced classes, making robust model training challenging. Collecting and annotating additional physiological data can be difficult and time-consuming, particularly for rare clinical events. Data augmentation can therefore be used to increase data diversity and improve model training.

- **Seminar:** Write a scoping review of data augmentation methods for physiological time-series classification, covering signal transformations, noise injection, window-based augmentation, and synthetic data generation across signals such as EEG, ECG, PPG, and respiration.

B PG	✗
M Sem	✓
M Lab	✗

- **Tools:** Scientific Research



Master Seminar & Lab

Missing Sensor Data in Multimodal Health Monitoring



AG Krüger (Nazila.AhmadiDaryakenari@ukbonn.de)

Multimodal health monitoring uses signals such as EEG, ECG, PPG, and respiration. In real-world recordings, some sensors may become noisy, disconnected, or temporarily unavailable. Therefore, models should remain reliable even when some sensor data are missing.

- **Seminar:** Write a scoping review of approaches for handling missing sensor data in multimodal physiological monitoring systems. Compare imputation, modality dropout, reconstruction, and models designed to operate with missing modalities.
- **Tools:** Scientific Research

B PPG	✗
M Sem	
✓	
M Lab	✗

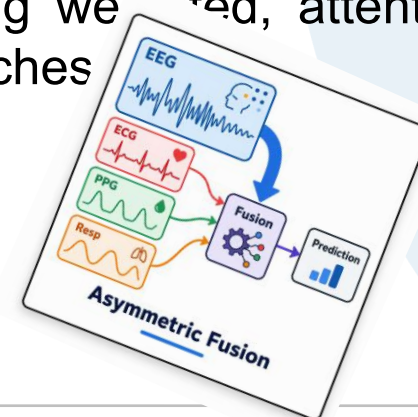


Asymmetric Fusion Strategies for Multimodal Biosignals

AG Krüger (Nazila.AhmadiDaryakenari@ukbonn.de)

Multimodal biosignals do not always contribute equally to a prediction task. One signal may provide the main information, while other modalities provide complementary information to improve robustness and decision-making.

- **Seminar:** Write a scoping review of fusion strategies where different biosignal modalities contribute unequally to the final prediction, including weighted, attention-based, gated, reliability-aware, and primary–auxiliary fusion approaches
- **Tools:** Scientific Research



B PG	✗
M Sem	✓
M Lab	✗



Master Seminar & Lab

Self-Supervised Learning for Unlabeled Physiological Time-Series

AG Krüger (Nazila.AhmadiDaryakenari@ukbonn.de)

Physiological recordings are often available in large amounts, but obtaining reliable labels requires expert annotation and can be time-consuming. Self-supervised learning allows models to learn useful representations directly from unlabeled signals.

- **Seminar:** Write a scoping review of self-supervised learning methods for unlabeled physiological time-series, including contrastive learning, masked signal modeling, predictive learning, and reconstruction-based approaches.
- **Tools:** Scientific Research



B PG	✗
M Sem	✓
M Lab	✗



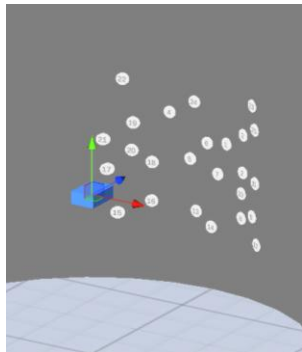
Master Seminar & Lab

Eye Tracking in a VR-Cognitive Evaluation Task

AG Krüger (melissa.steiningер@ukbonn.de, anna.jansen@ukbonn.de)



The EpiTrack is an established tool to measure cognitive impairment in people with epilepsy. An eye-tracked-VR version of one of the subtests, the TMT, was developed to get more detailed insights about people's cognition. For that, a robust, calibrated eye-tracking pipeline is necessary.



- **Research question:** *“How can eye tracking be implemented into Unity as a reusable pipeline that produces new insights about people’s cognition?”*
- **Lab:** Building on an already implemented VR-project: build a reusable Eye Tracking module, build calibration pipeline, perform a small study with healthy people, analyze results
- **Seminar:** Research about eye tracking and visual search in VR
- **Tools:** Unity, Pupil Labs Neon Eye Tracker

B PG	✗
M Sem	✓
M Lab	✓



Master Seminar & Lab

Getting at the bottom of the Uncanny Valley

AG Krüger (melissa.steininge@ukbonn.de)

The Uncanny Valley is a phenomenon that says that ~almost~ human like avatars elicit a feeling of eeriness. Can facial features, attractiveness and cuteness predict this feeling?



- **Research question:** “Can face features, attractiveness and cuteness predict eeriness?”
- **Lab:** Building on an already implemented VR-study: perform a small study, analyze by creating predictive models
- **Seminar:** Research about the Uncanny Valley, specifically in VR Avatars
- **Tools:** Unity, predictive modeling

B PG	✗
M Sem	✓
M Lab	✓



Master Seminar & Lab

Creating an Emotional VR Game: Rage.

AG Krüger (melissa.steining@ukbonn.de)



Games can elicit strong emotions through their mechanics and tasks. How can we deliberately use these design features to create an emotional experience?

→ also possible for other emotions!

- **Research question:** “What kind of design features in a game can elicit rage?”
- **Lab:** Design a VR game that elicits the emotion rage.
- **Seminar:** Research about eliciting rage in games.
- **Tools:** Unity, Virtual Reality

B PG	✗
M Sem	✓
M Lab	✓



Master Seminar & Lab

Handtracking Data Transformation and Visualization in Unity

AG Krüger (anna.jansen@ukbonn.de)

Transferring handtracking data from one skeleton representation into another can be challenging. Especially when thumb joint count differs, it can lead to unnatural movement in Unity.

- **Research question:** *“How can different hand skeleton representations be transformed to look natural in Unity visualization?”*
- **Lab:** Define transformation matrices for different hand skeleton representations suitable hand mesh and implement it in Unity.
- **Seminar:** Compare different hand models used in research and the different hand skeleton representations

B PG	✗
M Sem	✓
M Lab	✓

Tools: C#, handtracking data, Unity



Master Seminar & Lab

Universal Live-Viewer for Handtracking Data

AG Krüger (anna.jansen@ukbonn.de)

- **Research question:** *“How to implement a live-viewer which can take diverse handtracking data as input and visualize it together with a fitted hand mesh model?”*
- **Lab:** Implement the live-viewer and find a way to fit a hand mesh onto the joint data - you can use image data as an additional input to take handshape into consideration.
- **Seminar:** Compare different hand models used in research and the different hand skeleton representations
- **Tools:** Python

B PG	✗
M Sem	✗
M Lab	✓

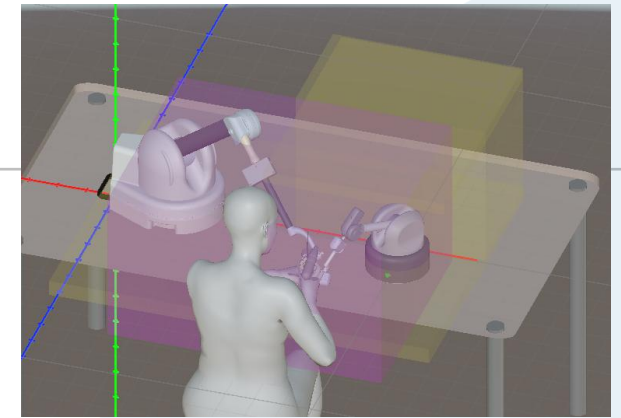


Master Seminar & Lab

Synthetic Handtracking Dataset

Creation and Evaluation

AG Krüger (anna.jansen@ukbonn.de)



Acquiring good training data is time consuming. Synthetic training data is a promising alternative but it needs to minimize domain gap (synthetic to real) and has to be well balanced.

- **Research question:** *“What does a good dataset need? What factors have an influence on model performance?”*
- **Lab:** Put together requirements for a well balanced dataset for a specific use case. Identify critical parameters that lead to a diverse dataset. Generate dataset.
Compare output of a prediction model with changing input.

B PG	✗
M Sem	✓
M Lab	✓

Seminar: Compare different synthetic hand datasets.

Tools: Python, existing Blender tool for data generation



Master Seminar & Lab

Robustness Evaluation of Camera-Based Heart Rate Measurements

AG Krüger (koray.simsek@ukbonn.de)



Camera-based heart rate estimation enables contactless monitoring using subtle color changes in the face. However, measurement quality strongly depends on environmental conditions, camera settings and subject-related factors.

- **Research question:** *“How robust is camera-based heart rate estimation under different environmental and subject-related conditions?”*
- **Lab:** Systematic evaluation of camera-based heart rate measurement under varying conditions; investigation of influencing factors such as lighting, indoor/outdoor scenarios, distance, motion, facial accessories and camera settings; comparison against a reference heart rate measurement; identification of the most relevant

B PG	✗
M Sem	✗
M Lab	✓

sources of error.



Master Seminar & Lab

Robustness Evaluation of Camera-Based Respiratory Rate Measurements

AG Krüger (koray.simsek@ukbonn.de)



Respiratory rate can be estimated contactlessly from depth or RGB camera data by detecting breathing-related body motion. Measurement performance depends on camera parameters, algorithms, hardware and recording conditions.

- **Research question:** *“Which parameters and algorithms provide robust camera-based respiratory rate estimation on resource-constrained hardware?”*
- **Lab:** Evaluation of respiratory rate estimation under varying recording conditions; investigation of frame rate, body pose, motion and clothing; comparison of different algorithms and software libraries; benchmarking of implementations on

B PG	✗
M Sem	✗
M Lab	✓

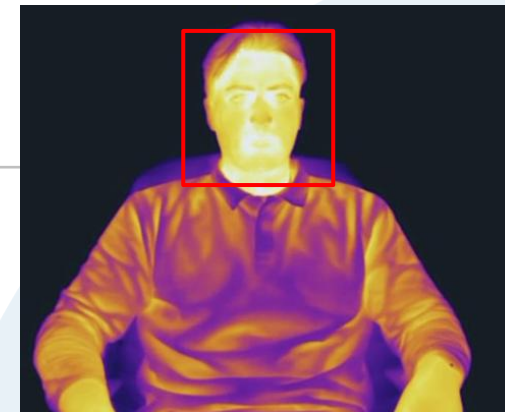
Jetson and OAK devices; comparison against a reference respiratory signal.

Tools: Python, OAK-D, Nvidia Jetson



Robust Face Tracking in Thermal Images

AG Krüger (koray.simsek@ukbonn.de)



Face detection in thermal images becomes challenging at larger distances due to low resolution and weak facial features.

- **Research question:** *“Which model provides the most robust face tracking in thermal images at larger distances?”*
- **Lab:** Comparison of face detection and tracking models; evaluation at different distances, angles and image resolutions; optimization for long-range thermal imaging.
- **Tools:** Python, Thermal camera, Nvidia Jetson

B PG	✗
M Sem	✗
M Lab	✓



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Master Seminar & Lab (Winter Term 2026/2027)

Prof. Dr. Björn Krüger // bkrueger@uni-bonn.de

Bonn, 06. Oktober 2026

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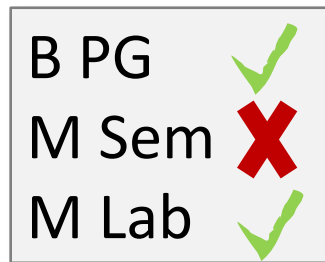
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Topic Selection

- Interested in a topic?
 - Contact the supervisor by e-mail or after the presentation!
 - E-mail addresses and names are provided on the slide(s) of the respective topic.
 - If in doubt, contact Jan Spindler (jspindle@cs.uni-bonn.de)



**These symbols tell you the
courses a topic is suitable for**

- Slides for organization and topics are available on our homepage:
 - <https://cg.cs.uni-bonn.de/teaching/ws-2026-27-ma-inf-2308-master-lab-graphics>
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